

Eikonal Equation as a Bridge to Schrödinger's Equation

Classical Limit and JWKB Approximation

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INTERNATIONAL YEAR OF
Quantum Science
and Technology



Outline

1. Main Idea
2. Historical Roots
3. JWKB Method
4. Conclusions and Future Work
5. References

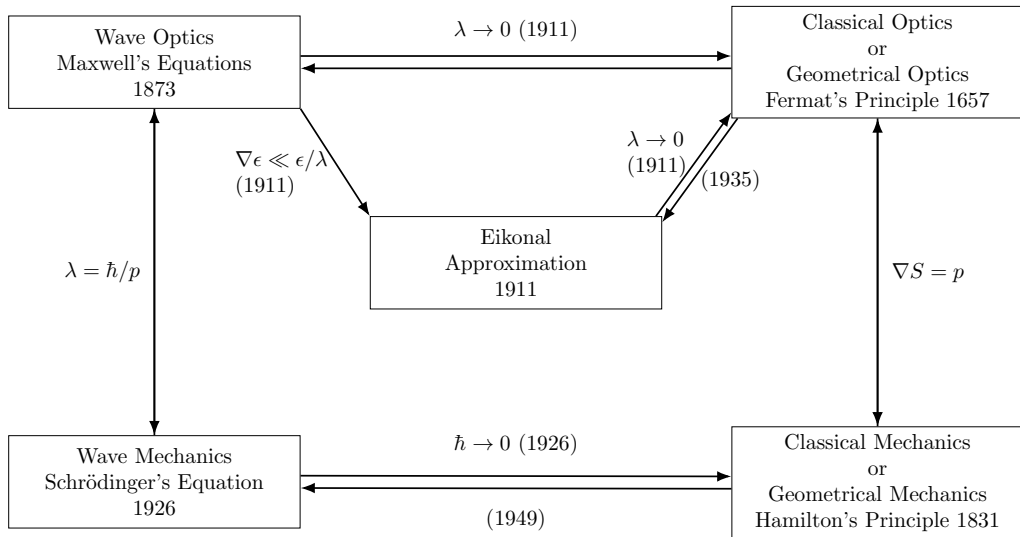


Figure 1: Map of equivalences. Adapted from Gilmore [12].

Main Idea

The Central Analogy

$$\frac{\text{Quantum Mechanics}}{\text{Classical Mechanics}} = \frac{\text{Wave Optics}}{\text{Geometric Optics}}$$

Quantum Mechanics is to Classical Mechanics as
Wave Optics is to Geometric Optics.

- $\lambda \rightarrow 0 \implies \text{Wave Opt.} \rightarrow \text{Geometric Opt.}$
- $\hbar \rightarrow 0 \implies \text{Quantum Mech.} \rightarrow \text{Classical Mech.}$

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Introduction: A Tale of Two Theories

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Historical Roots

Geometrical Optics

- Obeys Fermat's principle.
- Leads to the **Eikonal equation**:

$$(\nabla S)^2 = \frac{n^2}{c^2} \left(\frac{\partial S}{\partial t} \right)^2$$

Classical Mechanics

- Obeys the Principle of Least Action.
- Leads to the **Hamilton-Jacobi equation**:

$$(\nabla S)^2 = -2m \left(\frac{\partial S}{\partial t} + U \right)$$

Where S is the Classical Action.

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Hamilton's Analogy

The Eikonal and stationary Hamilton-Jacobi equations are formally identical.

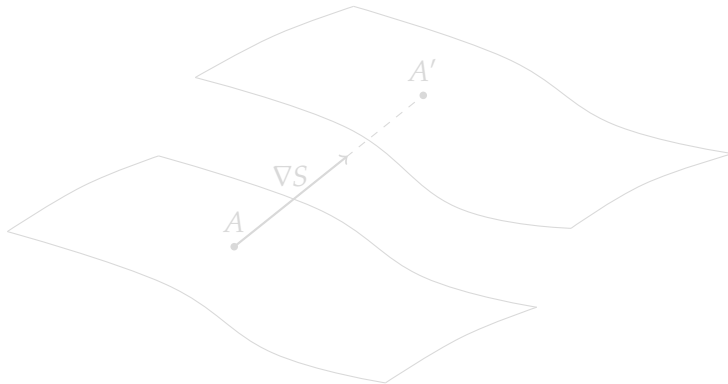


Figure 2: Displacement of a constant action surface after a time interval δt .

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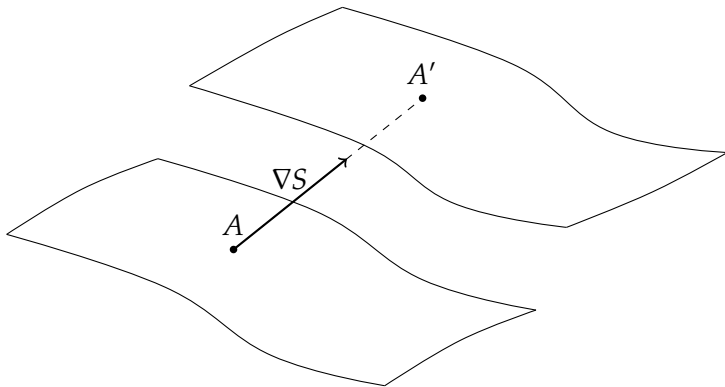
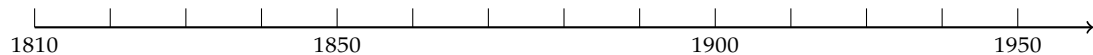


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JWKB Approximation – A Brief History



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Francesco Carlini (1789–1862)



Italian Astronomer

1810

1850

1950

JWKB Approximation – A Brief History

Carlini – (1817)



1810

1850

1900

1950

JWKB Approximation – A Brief History

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Joseph Liouville (1809–1882)



French Mathematician

George Green (1793–1841)



British Mathematician

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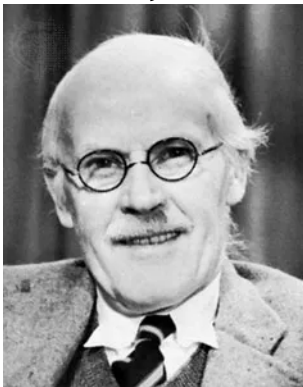
Carlini – (1817)



Liouville – (1837)



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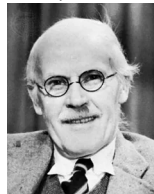
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JWKB Approximation – A Brief History

Liouville – (1837)

Gregor Wentzel (1898–1978)



German Physicist



Hans Kramers (1894–1952)



Dutch Physicist

Jeffreys – (1924)

Léon Brillouin (1889–1969)



French Physicist

JWKB Approximation – A Brief History

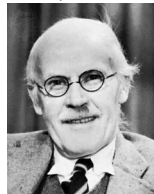
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Wentzel – (1926)



Kramers – (1926)



Brillouin – (1926)



JWKB Method

The Mathematical Setup

Problem: Perturbation of the Highest Derivative

Consider a second-order ODE where a small parameter $\epsilon \ll 1$ multiplies the highest derivative:

$$\epsilon^2 \frac{d^2 y}{dx^2} - Q(x)y = 0$$

This is a **singular perturbation problem**.

The JWKB Ansatz

We seek a solution of the form:

$$y(x) \sim \exp \left[\frac{1}{\epsilon} \sum_{n=0}^{\infty} \epsilon^n S_n(x) \right]$$

This form captures the rapid, oscillatory behavior dictated by the $1/\epsilon$ term.

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Recovering Classical Mechanics

1. **Start** with the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t) \right) \psi$$

2. Apply the JWKB ansatz, treating the phase S as the action:

$$\psi(\mathbf{r}, t) = A \exp \left(\frac{iS(\mathbf{r}, t)}{\hbar} \right)$$

3. Substitute the ansatz into the Schrödinger equation.

$$\frac{1}{2m} (\nabla S)^2 + V + \frac{\partial S}{\partial t} = \frac{i\hbar}{2m} \nabla^2 S$$

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$$\underbrace{\frac{1}{2m}(\nabla S)^2 + V}_{\text{Hamiltonian } H} + \frac{\partial S}{\partial t} = \underbrace{\frac{i\hbar}{2m}\nabla^2 S}_{\rightarrow 0 \text{ as } \hbar \rightarrow 0}$$

5 **Result:** We recover the classical Hamilton-Jacobi equation.

$$H(\mathbf{r}, \nabla S, t) + \frac{\partial S}{\partial t} = 0$$

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Interplay Between Maths and Physics

The Mathematical Condition

The first-order condition requires:

$$|\epsilon S_2(x)| \ll |S_1(x)|$$

The Physical Picture

Local de Broglie wavelength,
 $\lambda(x) = h/p(x)$:

$$\left| \frac{d\lambda(x)}{dx} \right| \ll 1$$

The Core Idea

The approximation holds when the potential $V(x)$ changes so slowly that the particle's momentum (and thus its wavelength) remains **almost constant** over the distance of a single wavelength.

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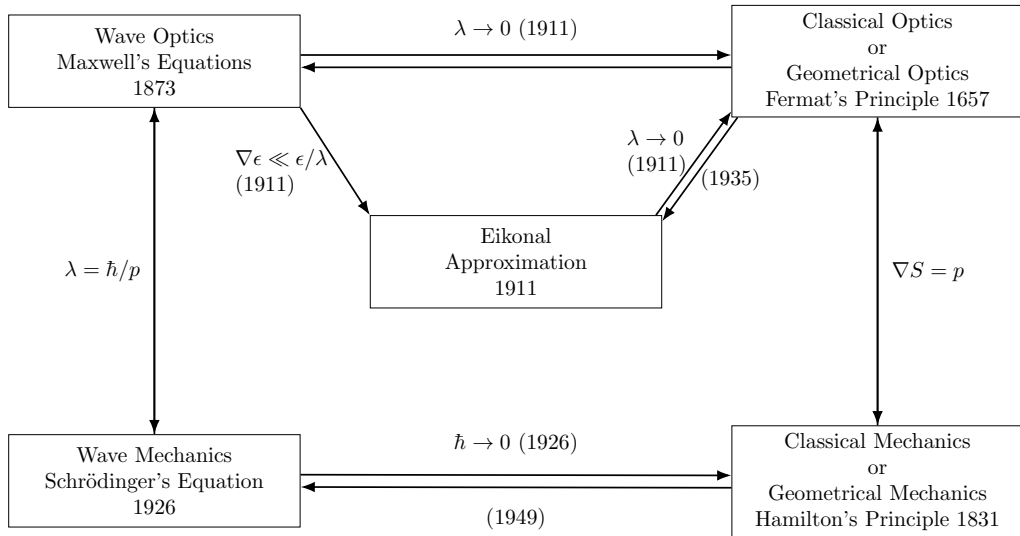


Figure 3: Map of equivalences. Adapted from Gilmore [12].

Conclusions and Future Work

The Central Takeaway

The formal identity between the **Eikonal equation** (Optics) and the **Hamilton-Jacobi equation** (Mechanics) is profound, not coincidental.

- This analogy guided de Broglie and Schrödinger to the **wave theory of matter**.
- The **JWKB approximation** shows classical trajectories emerging as the leading-order of QM in the limit $\hbar \rightarrow 0$.
- Classical Mechanics is the **geometrical optics** of the underlying wave mechanics (Schrödinger's Eq.).

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Geometric & Symplectic Methods

- H-J theory as a gateway to:
- **Symplectic & Contact Geometry**
- Goal: Coordinate-independent descriptions of dynamics.

Semi-classical Methods in QFT

- Apply to **Supersymmetric (SUSY) theories**.
- **SWKB** approximation can yield exact results.
- Powerful tool for non-perturbative effects.

Relativistic Semi-Classics

- Extend from the Schrödinger to the **Dirac equation**.
- Develop semi-classical propagators for spin-1/2 particles.
- Derive classical spin dynamics, e.g., the **BMT equation**.

References

References

- [1] M V Berry and K E Mount. **“Semiclassical approximations in wave mechanics”**. In: *Reports on Progress in Physics* 35.1 (Jan. 1, 1972), pp. 315–397. ISSN: 00344885. DOI: 10.1088/0034-4885/35/1/306. URL: <https://iopscience.iop.org/article/10.1088/0034-4885/35/1/306> (visited on 09/07/2025).
- [2] **Brillouin, Léon, 1889-1969**. URL: <https://history.aip.org/phn/11412022.html>.

- [3] J.D. Bulnes et al. **“From the Hamilton-Jacobi equation to the Schrödinger equation and vice versa, without additional terms and approximations”**. In: *Himalayan Physics* 11 (May 24, 2024), pp. 21–27. ISSN: 2542-2545. DOI: 10.3126/hp.v11i1.66159. URL: <https://www.nepjol.info/index.php/HP/article/view/66159> (visited on 09/07/2025).
- [4] Wikimedia Commons. ***File:Joseph liouville.jpeg — Wikimedia Commons, the free media repository.*** 2024. URL: %5Curl%7Bhttps://commons.wikimedia.org/w/index.php?title=File:Joseph_liouville.jpeg&oldid=915243666%7D.

- [5] Cosmos Magazine and Jeff Glorfeld. “England’s enigmatic mathematician is born”. In: (Dec. 9, 2020). URL: <https://cosmosmagazine.com/science/mathematics/this-week-in-science-history-englands-enigmatic-mathematician-is-born/>.
- [6] Darryl D Holm. *Geometric Mechanics - Part I: Dynamics And Symmetry (2nd Edition)*. OCLC: 1480897750. 2011. ISBN: 978-1-84816-776-6.
- [7] Robert B. Dingle and R. B. Dingle. *Asymptotic expansions: their derivation and interpretation*. London: Acad. Press, 1973. 521 pp. ISBN: 978-0-12-216550-4.
- [8] P. A. M. Dirac. *The principles of quantum mechanics*. 4. ed. (rev.); Repr. The @international series of monographs on physics. Oxford: Clarendon Pr., 1978. 314 pp. ISBN: 0-19-851208-2.

- [9] Paul Adrien Maurice Dirac. **“The quantum theory of the electron”**. In: *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character* 117.778 (Feb. 1928). Publisher: The Royal Society, pp. 610–624. ISSN: 2053-9150. DOI: [10.1098/rspa.1928.0023](https://doi.org/10.1098/rspa.1928.0023).
- [10] N. Fröman and P.O. Fröman. ***JWKB Approximation: Contributions to the Theory***. North-Holland Publishing Company; [sole distributors for U.S.A.: Interscience Publishers, New York], 1965. ISBN: 978-0-7204-0085-4. URL: <https://books.google.com.co/books?id=82U6AAAAMAAJ>.

- [11] Nanny Fröman and Per Olof Fröman. *Physical Problems Solved by the Phase-Integral Method*. 1st ed. Cambridge University Press, June 13, 2002. ISBN: 978-0-521-81209-2. DOI: 10.1017/CB09780511535086. URL: <https://www.cambridge.org/core/product/identifier/9780511535086/type/book> (visited on 10/05/2025).
- [12] R. Gilmore. *The Classical Origin of the Schrödinger Equations and the Darwin Correction Terms*.
- [13] *Hans Kramers*. URL: <https://prabook.com/web/hans.kramers/3771757>.
- [14] C.E. Liberati. *Ricordi del Terzo Congresso Scientifico Italiano ossia ritratti di trentasei fra i suoi componenti*. J. Grazzini, 1842. URL: <https://books.google.com.co/books?id=2GbQXPYYvhcC>.

- [15] Gunter Ludwig. *Wave Mechanics: Selected Reading in Physics*. Ed. by D. ter Haar. Burlington: Elsevier Science, 1968. 237 pp. ISBN: 978-0-08-203204-5.
- [16] Jaume Masoliver and Ana Ros. **“From classical to quantum mechanics through optics”**. In: *European Journal of Physics* 31.1 (Jan. 1, 2010), pp. 171–192. ISSN: 0143-0807, 1361-6404. DOI: [10.1088/0143-0807/31/1/016](https://doi.org/10.1088/0143-0807/31/1/016). URL: <https://iopscience.iop.org/article/10.1088/0143-0807/31/1/016> (visited on 09/07/2025).
- [17] E. Schrödinger. **“Quantisierung als Eigenwertproblem (I)”**. In: *Annalen der Physik* 384.4 (Jan. 1926). Publisher: Wiley, pp. 361–376. ISSN: 1521-3889. DOI: [10.1002/andp.19263840404](https://doi.org/10.1002/andp.19263840404).

- [18] E. Schrödinger. **“Quantisierung als Eigenwertproblem (II)”**. In: *Annalen der Physik* 384.6 (Jan. 1926). Publisher: Wiley, pp. 489–527. ISSN: 1521-3889. DOI: 10.1002/andp.19263840602.
- [19] E. Schrödinger. **“Quantisierung als Eigenwertproblem (III)”**. In: *Annalen der Physik* 385.13 (Jan. 1926). Publisher: Wiley, pp. 437–490. ISSN: 1521-3889. DOI: 10.1002/andp.19263851302.
- [20] E. Schrödinger. **“Quantisierung als Eigenwertproblem (IV)”**. In: *Annalen der Physik* 386.18 (Jan. 1926). Publisher: Wiley, pp. 109–139. ISSN: 1521-3889. DOI: 10.1002/andp.19263861802.

- [21] Lachezar S. Simeonov. *Derivation of Schrodinger's equation from the Hamilton-Jacobi and the Eikonal equations*. Sept. 21, 2024. DOI: 10.48550/arXiv.2410.03689. arXiv: 2410.03689[quant-ph]. URL: <http://arxiv.org/abs/2410.03689> (visited on 09/07/2025).
- [22] The Editors of Encyclopaedia Britannica. *Sir Harold Jeffreys | Astronomy, Geophysics, Mathematical Physics*. July 20, 1998. URL: <https://www.britannica.com/biography/Harold-Jeffreys>.
- [23] *Wentzel, Gregor, 1898-1978*. URL: <https://history.aip.org/phn/11611006.html>.
- [24] Wolfgang Werner. *Hamiltonsche Mechanik und Quantenmechanik*. Wiesbaden: Springer Vieweg, 2023. 1670 pp. ISBN: 978-3-658-41538-9.

- [25] Westminster Abbey. *George Green / Westminster Abbey*. URL: <https://www.westminster-abbey.org/abbey-commemorations/commemorations/george-green> (visited on 10/01/2025).
- [26] Roscoe B. White. *Asymptotic analysis of differential equations*. Revised Edition. London: Imperial College Press, 2010. 1 p. ISBN: 978-1-911298-59-5. DOI: 10.1142/p735.

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Questions or Comments?

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